Exercise J-M

1. The value of $\sum_{r=0}^{22} {}^{22}C_r \, {}^{23}C_r$ is : **[JEE(Main)Jan.2023]**

$\sum_{r=0}^{22} {}^{22}C_r \, {}^{23}C_r$ का मान है : **[JEE(Main)Jan.2023]**

(A) ${}^{44}C_{22}$ (B) ${}^{45}C_{24}$ (C) ${}^{45}C_{23}$ (D) ${}^{44}C_{23}$

Ans. C

Sol. $\sum_{r=0}^{22} {}^{22}C_r \, {}^{23}\sum_{r=0}^{22} {}^{22}C_r \, {}^{23}C_r = \sum_{r=0}^{22} {}^{22}C_r \, {}^{23}\sum_{r=0}^{22} {}^{22}C_r \, {}^{23}C_{23-r}$

$= {}^{22}C_0 \, {}^{23}C_{23} + {}^{22}C_1 \, {}^{23}C_{22} + \cdots + {}^{22}C_{21} \, {}^{23}C_2 + {}^{22}C_{22} \, {}^{23}C_1$

$(1+x)^{22} = {}^{22}C_0 + {}^{22}C_1 x + \cdots + {}^{22}C_{21} x^{21} + {}^{22}C_{22} x^{22}$

$(1+x)^{23} = {}^{23}C_0 + {}^{23}C_1 x + \cdots + {}^{23}C_{22} x^{22} + {}^{23}C_{23} x^{23}$

coefficient of x^{23} in $(1+x)^{22}(1+x)^{23} = \sum_{r=0}^{22} {}^{22}C_r \, {}^{23}\sum_{r=0}^{22} {}^{22}C_r \, {}^{23}C_{23-r}$

$= {}^{45}C_{23}$

2. If $({}^{30}C_1)^2 + 2({}^{30}C_2)^2 + 3({}^{30}C_3)^2 + \ldots\ldots + 30({}^{30}C_{30})^2 = \frac{\alpha 60!}{(30!)^2}$ then α is equal to :

[JEE(Main)Jan.2023]

यदि $({}^{30}C_1)^2 + 2({}^{30}C_2)^2 + 3({}^{30}C_3)^2 + \ldots\ldots + 30({}^{30}C_{30})^2 = \frac{\alpha 60!}{(30!)^2}$ है, तो α बराबर है :

[JEE(Main)Jan.2023]

(A) 15 (B) 60 (C) 10 (D) 30

Ans. A

Sol. $S = 0 \cdot \left({}^{30}C_0\right)^2 + 1 \cdot \left({}^{30}C_1\right)^2 + 2 \cdot \left({}^{30}C_2\right)^2 + \ldots\ldots + 30 \cdot \left({}^{30}C_{30}\right)^2$

$S = 30 \cdot \left({}^{30}C_0\right)^2 + 29 \cdot \left({}^{30}C_1\right)^2 + 28 \cdot \left({}^{30}C_2\right)^2 + \ldots + 0 \cdot \left({}^{30}C_0\right)^2$

$2S = 30 \cdot \left({}^{30}C_0{}^2 + {}^{30}C_1{}^2 + \ldots\ldots \cdot + {}^{30}C_{30}{}^2\right)$

$S = 15 \cdot {}^{60}C_{30} = 15 \cdot \frac{60!}{(30!)^2}$

$\frac{15 \cdot 10!}{(30!)^2} = \frac{\alpha \cdot 60!}{(30!)^2}$

$\Rightarrow \alpha = 15$

3. If a_r is the coefficient of x^{10-r} in the Binomial expansion of $(1+x)^{10}$, then $\sum_{r=1}^{10} r^3\left(\frac{a_r}{a_{r-1}}\right)^2$ is equal to :

[JEE(Main)Jan.2023]

यदि $(1+x)^{10}$ के द्विपद प्रसार में x^{10-r} का गुणांक a_r है, तो $\sum_{r=1}^{10} r^3\left(\frac{a_r}{a_{r-1}}\right)^2$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 4895 (B) 5445 (C) 3025 (D) 1210

Ans. D

Sol. Coeff. of x^{10-r} in $(1+x)^{10}$

$a_r = {}^{10}C_r$

$$\frac{a_r}{a_{r-1}} = \frac{{}^{10}C_r}{{}^{10}C_{r-1}} = \frac{10-r+1}{r} = \frac{11-r}{r}$$

$$\sum_{r=0}^{10} r^3\left(\frac{11-r}{r}\right)^2 = \sum_{r=0}^{10} r(11-r)^2$$

$$= \sum_{r=0}^{10}\left(r^3 - 22r^2 + 121r\right)$$

$$= \sum_{r=0}^{10} r^3 - 22\sum_{r=0}^{10} r^2 + 121\sum_{r=0}^{10} r = \left(\frac{10(10+1)}{2}\right)^2 - 22\times\frac{10(10+1)(20+1)}{6} + 121\times\frac{10\cdot 11}{2}$$

$$= 55^2 - 11\times 11\times 70 + 5\times 11^3 = 11^2[25-70+55] = 11^2\times 10 = 1210$$

4. $\sum_{k=0}^{6} {}^{51-k}C_3$ is equal to : **[JEE(Main)Jan.2023]**

$\sum_{k=0}^{6} {}^{51-k}C_3$ बराबर है : **[JEE(Main)Jan.2023]**

(A) ${}^{51}C_4 - {}^{45}C_4$ (B) ${}^{52}C_4 - {}^{45}C_4$ (C) ${}^{51}C_3 - {}^{45}C_3$ (D) ${}^{52}C_3 - {}^{45}C_3$

Ans. B

Sol. ${}^{51}C_3 + {}^{50}C_3 + ---------- + {}^{45}C_3$

$\Rightarrow {}^{45}C_4 + {}^{45}C_3 + {}^{46}C_3 + ---------- + {}^{51}C_3 - {}^{45}C_4$

$\Rightarrow {}^{46}C_4 + {}^{46}C_3 + --------$

$= {}^{51}C_4 + {}^{51}C_3 - {}^{45}C_4 = {}^{52}C_4 - {}^{45}C_4$

5. Let K be the sum of the coefficients of the odd powers of x in the expansion of $(1+x)^{99}$. Let a be the middle term in the expansion of $\left(2+\frac{1}{\sqrt{2}}\right)^{200}$. If $\frac{{}^{200}C_{99}K}{a} = \frac{2^\ell m}{n}$, where m and n are odd numbers, then the ordered pair (l, n) is equal to : **[JEE(Main)Jan.2023]**

माना $(1+x)^{99}$ के प्रसार में x की विषम घातों के गुणांको का योग K है। माना $\left(2+\frac{1}{\sqrt{2}}\right)^{200}$ के प्रसार में मध्य पद a है। यदि $\frac{{}^{200}C_{99}\,K}{a}=\frac{2^{\ell}\,m}{n}$ है, जहाँ *m* तथा *n* विषम संख्याऐं है, तो क्रमित युग्म (*l*, n) बराबर है। **[JEE(Main)Jan.2023]**

(A) (50, 51) (B) (51,101) (C) (50,101) (D) (51,99)

Ans. C

Sol. Sum of odd coeff. of $x=\frac{2^{99}-0}{2}=k$

$k=2^{98}$

Middle term $\alpha = T101 = {}^{200}C_{100}\frac{(2)^{100}}{2^{50}}$

$= {}^{200}C_{100}\,2^{50}$

Now $\frac{{}^{200}C^{99}\times k}{\alpha}=\frac{200!}{100!99!}\times\frac{2^{98}}{{}^{200}C_{100}\times 2^{50}}$

$=\frac{100}{101}\times 2^{48}$

$\frac{25}{101}\times 2^{50}$

6. If the coefficient of x^{15} in the expansion of $\left(ax^3+\frac{1}{bx^{1/3}}\right)^{15}$ is equal to the coefficient of x^{-15} in the expansion of $\left(ax^{1/3}-\frac{1}{bx^{1/3}}\right)^{15}$, where a and b are positive real numbers, then for each such ordered pair (a, b) :

[JEE(Main)Jan.2023]

यदि $\left(ax^3+\frac{1}{bx^{1/3}}\right)^{15}$ के प्रसार में x^{15} का गुणांक, $\left(ax^{1/3}-\frac{1}{bx^{1/3}}\right)^{15}$, के प्रसार में x^{-15} के गुणांक के बराबर है, जहाँ a तथा b घनात्मक संख्याएँ है, तो ऐसे प्रत्येक क्रमित युग्म (a, b) के लिए : **[JEE(Main)Jan.2023]**

(A) a = b (B) ab = 1 (C) a = 3b (D) ab = 3

Ans. B

Sol. Coefficient of x^{15} in $\left(ax^3+\frac{1}{bx^{1/3}}\right)^{15}$

$T_{r+1}={}^{15}C_r\left(ax^3\right)^{15-r}\left(\frac{1}{bx^{1/3}}\right)^r$

$45 - 3r - \frac{r}{3} = 15$

$30 = \frac{10r}{3}$

$r = 9$

Coefficient of $x^{15} = {}^{15}C_9\, a^6\, b^{-9}$

Coefficient of x^{-15} in $\left(ax^{1/3} - \frac{1}{bx^3}\right)^{15}$

$T_{r+1} = {}^{15}C_r \left(ax^{1/3}\right)^{15-r} \left(-\frac{1}{bx^3}\right)^r$

$5 - \frac{r}{3} - 3r = -15$

$\frac{10r}{3} = 20$

$r = 6$

Coefficient $= {}^{15}C_6\, a^9 \times b^{-6}$

$\Rightarrow \frac{a^9}{b^6} = \frac{a^6}{b^9}$

$\Rightarrow a^3b^3 = 1 \Rightarrow ab = 1$

7. The coefficient of x^{301} in $(1+x)^{500} + x(1+x)^{499} + x^2(1+x)^{498} + \ldots\ldots + x^{500}$ is :

[JEE(Main)Jan.2023]

$(1+x)^{500} + x(1+x)^{499} + x^2(1+x)^{498} + \ldots\ldots + x^{500}$ में x^{301} का गुणांक है : **[JEE(Main)Jan.2023]**

(A) ${}^{501}C_{302}$ (B) ${}^{500}C_{300}$ (C) ${}^{500}C_{301}$ (D) ${}^{501}C_{200}$

Ans. D

Sol. $(1+x)^{500} + x(1+x)^{499} + x^2(1+x)^{498} + \ldots + x^{500}$

$$= (1+x)^{500} \left\{ \frac{1 - \left(\frac{x}{1+x}\right)^{501}}{1 - \frac{x}{1+x}} \right\}$$

$$= (1+x)^{500} \frac{\left((1+x)^{501} - x^{501}\right)}{(1+x)^{501}} \cdot (1+x)$$

$= (1+x)^{501} - x^{501}$

Coefficient of x^{301} in $(1+x)^{501} - x^{501}$ is given by ${}^{501}C_{301} = {}^{501}C_{200}$

8. Let $x=\left(8\sqrt{3}+13\right)^{13}$ and $y=\left(7\sqrt{2}+9\right)^{9}$. If [t] denotes the greatest integer $\le t$, then :

[JEE(Main)Jan.2023]

(A) [x] is even but [y] is odd (B) [x] + [y] is even

(C) [x] and [y] are both odd (D) [x] is odd but [y] is even

माना $x=\left(8\sqrt{3}+13\right)^{13}$ और $y=\left(7\sqrt{2}+9\right)^{9}$ है। यदि [t] महत्तम पूर्णांक $\le t$ है, तब : **[JEE(Main)Jan.2023]**

(A) [x] सम है परन्तु [y] विषम है (B) [x] + [y] सम है

(C) [x] एवं [y] दोनों विषम है (D) [x] विषम है परन्तु [y] सम है

Ans. B

Sol. $x=\left(8\sqrt{3}+13\right)={}^{13}C_0\cdot\left(8\sqrt{3}\right)^{13}+{}^{13}C_1\left(8\sqrt{3}\right)^{12}+....$

$x'=\left(8\sqrt{3}-13\right)^{13}={}^{13}C_0\left(8\sqrt{3}\right)^{13}-{}^{13}C_1\left(8\sqrt{3}\right)^{12}(13)^1+...$

$x-x'=2\left[{}^{13}C_1\cdot\left(8\sqrt{3}\right)^{12}(13)^1+{}^{13}C_3\left(8\sqrt{3}\right)^{10}\cdot(13)^3....\right]$

therefore, $x-x'$ is even integer, hence [x] is evem

Now, $y=\left(7\sqrt{2}+9\right)^{9}={}^{9}C_0\left(7\sqrt{2}\right)^{9}+{}^{9}C_1\left(7\sqrt{2}\right)^{8}(9)^1+{}^{9}C_2\left(7\sqrt{2}\right)^{7}(9)^2.....$

$y'=\left(7\sqrt{2}-9\right)^{9}={}^{9}C_0\left(7\sqrt{2}\right)^{9}-{}^{9}C_1\left(7\sqrt{2}\right)^{8}(9)^1+{}^{9}C_2\left(7\sqrt{2}\right)^{7}(9)^2.....$

$y-y'=2\left[{}^{9}C_1\left(7\sqrt{2}\right)^{8}(9)^1+{}^{9}C_3\left(7\sqrt{2}\right)^{6}(9)^3+...\right]$

$y-y'=$ Even integer, hence [y] is even

9. The value of $\frac{1}{1!50!}+\frac{1}{3!48!}+\frac{1}{5!46!}+.....+\frac{1}{49!2!}+\frac{1}{51!1!}$ is : **[JEE(Main)Jan.2023]**

$\frac{1}{1!50!}+\frac{1}{3!48!}+\frac{1}{5!46!}+.....+\frac{1}{49!2!}+\frac{1}{51!1!}$ का मान है : **[JEE(Main)Jan.2023]**

(A) $\frac{2^{51}}{51!}$ (B) $\frac{2^{50}}{51!}$ (C) $\frac{2^{51}}{50!}$ (D) $\frac{2^{50}}{50!}$

Ans. B

Sol. $\sum_{r=1}^{26}\frac{1}{(2r-1)!(51-(2r-1))!}=\sum_{r=1}^{26}{}^{51}C_{(2r-1)}\frac{1}{51!}$

$=\frac{1}{51!}\left\{{}^{51}C_1+{}^{51}C_3+...+{}^{51}C_{51}\right\}=\frac{1}{51!}\left(2^{50}\right)$

10. The sum $\sum_{n=1}^{\infty}\frac{2n^2+3n+4}{(2n)!}$ is equal to : **[JEE(Main)Jan.2023]**

योगफल $\sum_{n=1}^{\infty}\frac{2n^2+3n+4}{(2n)!}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{13e}{4}+\frac{5}{4e}$ (B) $\frac{13e}{4}+\frac{5}{4e}-4$

(C) $\frac{11e}{2}+\frac{7}{2e}$ (D) $\frac{11e}{2}+\frac{7}{2e}-4$

Ans. B

Sol. $\sum_{n=1}^{\infty}\frac{2n^2+3n+4}{2n!}$

$=\frac{1}{2}\sum_{n=1}^{\infty}\frac{2n(2n-1)+8n+8}{2n!}$

$=\frac{1}{2}\sum_{n=1}^{\infty}\left\{\frac{1}{(2n-2)!}+\frac{4}{(2n-1)!}+\frac{8}{2n!}\right\}$

$=\frac{1}{2}\left\{\left(\frac{e+e^{-1}}{2}\right)+4\left(\frac{e-e^{-1}}{2}\right)+8\left(\frac{e+e^{-1}}{2}-1\right)\right\}$

$=\frac{1}{4}\left(13e+5e^{-1}-16\right)$

$=\frac{13}{4}e+\frac{5}{4e}-4$

11. The coefficient of x^{-6}, in the expansion of $\left(\frac{4x}{5}+\frac{5}{2x^2}\right)^9$, is _____. **[JEE(Main)Jan.2023]**

$\left(\frac{4x}{5}+\frac{5}{2x^2}\right)^9$ के प्रसार में x^{-6} का गुणांक है। **[JEE(Main)Jan.2023]**

Ans. 5040

Sol. $T_{r+1}={}^9C_r\left(\frac{5}{2x^2}\right)^r\cdot\left(\frac{4x}{5}\right)^{9-r}={}^9C_r\frac{2^{18-3r}}{5^{9-2r}}x^{9-3r}$

for coefficient of $x^{-6}\Rightarrow 9-3r=-6\Rightarrow r=5$

coff. of $x^{-6}=\frac{{}^9C_5.2^3}{5^{-1}}=\frac{9.8.7.5}{4.3.2.1}.(8\times5)$

$= 5040$

12. If the constant term in the binomial expansion of $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^{\ell}}\right)^9$ is –84 and the coefficient of x^{-3l} is $2^{\alpha}\beta$, where $\beta<0$ is an odd number, then $|\alpha\ell-\beta|$ is equal to _____. **[JEE(Main)Jan.2023]**

यदि $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^{\ell}}\right)^9$ के द्विपद प्रसार में अचर पद –84 है तथा x^{-3l} का गुणांक $2^{\alpha}\beta$ है, जहाँ $\beta<0$ एक विषम संख्या है, तो $|\alpha\ell-\beta|$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 98

Sol. In, $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^{\ell}}\right)^9$

$$T_{r+1} = {}^9C_r \frac{\left(x^{5/2}\right)^{9-r}}{2^{9-r}}\left(\frac{-4}{x^{\ell}}\right)^r$$

$$= (-1)^r \frac{{}^9C_r}{2^{9-r}} 4^r x^{\frac{45}{2}-\frac{5r}{2}-lr}$$

$$= 45-5r-21r = 0$$

$$r = \frac{45}{5+21} \qquad \text{_____(1)}$$

Now, according to the question, $(-1)^r \frac{{}^9C_r}{2^{9-r}} 4^r = -84$

$$= (-1)^r \, {}^9C_r \, 2^{3r-9} = 21\times 4$$

Only natural value of r possible if $3r-9=0$

$r = 3$ and ${}^9C_3 = 84$

$\therefore l = 5$ from equation (1)

Now, coefficient of $x^{-31} = x^{\frac{45}{2}-\frac{5r}{2}-lr}$ at $l = 5$, gives

$r = 5$

$$\therefore {}^9C_5(-1)\frac{4^5}{2^4} = 2^{\alpha}\times\beta$$

$$= -63\times 2^7$$

$\Rightarrow \alpha = 7, \beta = -63$

$\therefore$ value of $|\alpha\ell - \beta| = 98$

13. Suppose $\sum_{r=0}^{2023} r^2 \, {}^{2023}C_r = 2023 \times \alpha \times 2^{2022}$. Then the value of α is _____. **[JEE(Main)Jan.2023]**

माना $\sum_{r=0}^{2023} r^2 \, {}^{2023}C_r = 2023 \times \alpha \times 2^{2022}$ है। तो α का मान है _____ **[JEE(Main)Jan.2023]**

Ans. 1012

Sol. $\sum r \cdot {}^{2023}C_r = \sum r \cdot (2023)^{2022} C_{r-1}$

$2023 \sum r^{2022} C_{r-1}$

$2023\left(\sum_{1}^{2023} (r-1)^{2022} C_{r-1} + {}^{2022}C_{r-1}\right)$

$= 2023\left(\left(\sum_{r=2}^{2023} (2022)^{2021} C_{r-2}\right) + \sum_{r=1}^{2023} {}^{2022}C_{r-1}\right) = 2023\left(2022 \cdot 2^{2021} + 2^{2022}\right)$

$= 2^{2022} \cdot 2023(1011+1) = 2^{2022}(1012) \cdot (2023)$

$\because \alpha = 1012$

14. Let the sum of the coefficients of the first three terms in the expansion of $\left(x - \frac{3}{x^2}\right)^n$, $x \neq 0, n \in N$, be 376. Then the coefficients of x^4 is _____. **[JEE(Main)Jan.2023]**

माना $\left(x - \frac{3}{x^2}\right)^n$, $x \neq 0, n \in N$ के प्रसार में प्रथम तीन पदों के गुणांकों का योग 376 है। तो x^4 का गुणांक _____ है।

[JEE(Main)Jan.2023]

Ans. 406

Sol. ${}^nC_0 - {}^nC_1 \cdot 3 + {}^nC_2 \cdot 3^2 = 376$

$\Rightarrow n = 10$

Now $T_{r+1} = {}^{10}C_r \cdot \left(\frac{-3}{x^2}\right)^r \cdot (x)^{10-r}$

$= {}^{10}C_r \cdot (-3)^r \cdot x^{10-3r}$

For $x^4 \Rightarrow 10 - 3r = 4 \Rightarrow r = 2$

Coff. of $x^4 = {}^{10}C_2 \cdot (-3)^2 = 405$

15. The constant term in the expansion of $\left(2x+\frac{1}{x^7}+3x^2\right)^5$ is _____. **[JEE(Main)Jan.2023]**

$\left(2x+\frac{1}{x^7}+3x^2\right)^5$ के प्रसार में अचर पद है। **[JEE(Main)Jan.2023]**

Ans. 1080

Sol. General term is $\sum\frac{5!(2x)^{n_1}\left(x^{-7}\right)^{n_2}\left(3x^2\right)^{n_3}}{n_1!n_2!n_3!}$

For constant term,

$n_1 + 2n_3 = 7n_2$

& $n_1 + n_2 + n_3 = 5$

Only possibility $n_1 = 1, n_2 = 1, n_3 = 3$

$\Rightarrow$ constant term = 1080

16. The remainder when $(2023)^{2023}$ is divided by 35 is _____. **[JEE(Main)Jan.2023]**

$(2023)^{2023}$ को 35 से विभाजित करने पर शेषफल है। **[JEE(Main)Jan.2023]**

Ans. 7

Sol. $-72023 = -7 \times 7^{2022}$

$$=\frac{-7^{2022}}{5}$$

$$=\frac{-[50-1]^{1011}}{5}=\frac{[5\lambda-1]}{5}$$

remainder = + 1

when divided by 5 now remainder is 7 when divided 35

17. If the co-efficient of x^9 in $\left(\alpha x^3+\frac{1}{\beta x}\right)^{11}$ and the co-efficient of x^{-9} in $\left(\alpha x-\frac{1}{\beta x^3}\right)^{11}$ are equal, then $(\alpha\beta)^2$ is equal to _____. **[JEE(Main)Jan.2023]**

यदि $\left(\alpha x^3+\frac{1}{\beta x}\right)^{11}$ के प्रसार में x^9 का गुणांक एवं $\left(\alpha x-\frac{1}{\beta x^3}\right)^{11}$ के प्रसार में x^{-9} का गुणांक बराबर हैं, तब $(\alpha\beta)^2$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 1

Sol. Coefficient of x^9 in $\left(\alpha x^3+\frac{1}{\beta x}\right)={}^{11}C_6\cdot\frac{\alpha^5}{\beta^6}$

$\because$ Both are equal

$$\therefore \frac{11}{C_6}\cdot\frac{\alpha^5}{\beta^6} = -\frac{11}{C_5}\cdot\frac{\alpha^6}{\beta^5}$$

$$\Rightarrow \frac{1}{\beta} = -\alpha$$

$$\Rightarrow \alpha\beta = -1$$

$$\Rightarrow (\alpha\beta)^2 = 1$$

18. Let the coefficients of three consecutive terms in the binomial expansion of $(1 + 2x)^n$ be in the ratio 2 : 5 : 8. Then the coefficient of the term, which is in the middle of these three terms, is _____ .

[JEE(Main)Jan.2023]

माना $(1 + 2x)^n$ द्विपद प्रसार में तीन क्रमागत पदों के गुणांकों का अनुपात 2 : 5 : 8 है। इन तीन पदों में मध्य पद का गुणांक है।

[JEE(Main)Jan.2023]

Ans. 1120

Sol. $t_{r+1} = {}^nC_r\,(2x)^r$

$$\Rightarrow \frac{{}^nC_r(2)^{r-1}}{{}^nC_r(2)^r} = \frac{2}{5}$$

$$\Rightarrow \frac{\frac{n!}{(r-1)!(n-r+1)!}}{\frac{n!(2)}{r!(n-r)!}} = \frac{2}{5}$$

$$\Rightarrow \frac{r}{n-r+1} = \frac{4}{5} \Rightarrow 5r = 4n - 4r + 4$$

$$\Rightarrow 9r = 4\,(n + 1) \qquad ....(1)$$

$$\Rightarrow \frac{{}^nC_r(2)^r}{{}^nC_{r+1}(2)^{r+1}} = \frac{5}{8}$$

$$\Rightarrow \frac{\frac{n!}{r!(n-r)!}}{\frac{n!}{(r+1)!(n-r-1)!}} = \frac{5}{4} \Rightarrow \frac{r+1}{n-r} = \frac{5}{4}$$

$$\Rightarrow 4r + 4 = 5n - 5r \Rightarrow 5n - 4 = 9r \qquad ....(2)$$

From (1) and (2)

$$\Rightarrow 4n + 4 = 5n - 4 \Rightarrow n = 8$$

$$(A) \Rightarrow r = 4$$

so, coefficient of middle term is

$${}^{8}C_4 2^4 = 16 \times \frac{8\times7\times6\times5}{4\times3\times2\times1} = 16\times70 = 1120$$

19. Let $\sum_{n=0}^{\infty}\frac{n^3((2n)!)+(2n-1)(n!)}{(n!)((2n!))} = ae+\frac{b}{e}+c$, where $a,b,c\in Z$ and $e=\sum_{n=0}^{\infty}\frac{1}{n!}$. Then a^2-b+c is equal to :

[JEE(Main)Jan.2023]

माना $\sum_{n=0}^{\infty}\frac{n^3((2n)!)+(2n-1)(n!)}{(n!)((2n!))} = ae+\frac{b}{e}+c$, है जहाँ $a,b,c\in Z$ तथा $e=\sum_{n=0}^{\infty}\frac{1}{n!}$ है। तो a^2-b+c बराबर है।

[JEE(Main)Jan.2023]

Ans. 26

Sol. $$\sum_{n=0}^{\infty}\frac{n^3(2n)!}{n!(2n)!}+\sum_{n=0}^{\infty}\frac{(2n-1)n!}{n!(2n)!}$$

$$\sum_{n=0}^{\infty}\frac{(2n-1)n!}{n!(2n)!}$$

$$\sum_{n=0}^{\infty}\frac{(2n)\times n!}{n!\times(2n)!}-\frac{n!}{n!(2n)!}$$

$$\sum_{n=0}^{\infty}\frac{1}{(2n-1)!}-\sum_{n=0}^{\infty}\frac{1}{(2n)!}$$

$$\frac{e^x-e^{-x}}{2}\Bigg|_{x=1}-\frac{e^x+e^{-x}}{2}\Bigg|_{x=1}$$

$$=\sum_{n=0}^{\infty}\frac{n^3(2n)!}{n!(2n)!}=\sum_{n=0}^{\infty}\frac{n^3}{n!}$$

$n^3 = an(n-1)(n-2)+bn(n-1)+cn+d$

$n=1 \Rightarrow c=1$

$n=2 \Rightarrow 8=2b+2 \Rightarrow b=3$

$$=\sum_{n=0}^{\infty}\frac{n(n-1)(n-2)+3n(n-1)+n}{n!}$$

$$\sum_{n=3}^{\infty}\frac{1}{(n-3)!}+3\sum_{n=2}^{\infty}\frac{1}{(n-2)!}+\sum_{n=1}^{\infty}\frac{1}{(n-1)!}$$

$$=\sum_{n=0}^{\infty}\frac{n^3(2n)!}{n!(2n)!}+\sum_{n=0}^{\infty}\frac{(2n-1)n!}{n!(2n)!}$$

$$=5e+\frac{e}{2}-\frac{1}{e}-\frac{e}{2}+\frac{1}{e}$$

$= 5e - \frac{1}{e}$

$a = 5, b = -1, c = 0$

Now $a^2 - b + c = 26$

20. 50th root of a number x is 12 and 50th root of another number y is 18. Then the remainder obtained on dividing (x + y) by 25 is _____ . **[JEE(Main)Jan.2023]**

एक संख्या x का 50वाँ मूल 12 है एवं दूसरी संख्या y का 50वाँ मूल 18 है। तब (x + y) को 25 से विभाजित करने पर प्राप्त शेषफल है। **[JEE(Main)Jan.2023]**

Ans. 23

Sol. $(x)^{\frac{1}{50}} = 12$ and $(y)^{\frac{1}{50}} = 18$

$\Rightarrow x = 12^{50} \Rightarrow y = 18^{50}$

$x + y = 12^{50} + 18^{50}$

$(2^2 \times 3^1)^{50} + (3^2 \times 2)^{50}$

$= 6^{50} (2^{50} + 3^{50})$

$= (5 + 1)^{50} (4^{25} + 9^{25})$

$= (25\lambda + 1) [(5 - 1)^{25} + (10 - 1)^{25}]$

$= (25\lambda + 1) (25k - 1 + 25\mu - 1)$

$= (25\lambda + 1) (25t + 23)$

$= 25\lambda + 23$

so remiander is 23, when $x + y$ is divided by 25.

21. Let $\alpha > 0$, be the smallest number such that the expansion of $\left(x^{\frac{2}{3}} + \frac{2}{x^3} \right)^{30}$ has term $\beta x^{-\alpha}$, $\beta \in N$. Then α is equal to _____ . **[JEE(Main)Jan.2023]**

माना $\alpha > 0$ न्यूनतम संख्या है, जिसके लिए $\left(x^{\frac{2}{3}} + \frac{2}{x^3} \right)^{30}$ के प्रसार के एक पद में $\beta x^{-\alpha}$ तो α बराबर है _____ , $\beta \in N$

[JEE(Main)Jan.2023]

Ans. 2

Sol. $\alpha > 0, \left(x^{2/3} + \frac{2}{x^3} \right)^{30}$

$T_{r+1} = {}^{30}C_r \left(x^{2/3} \right)^{30-r} \left(\frac{2}{x^3} \right)^r$

$= {}^{30}C_r x^{20-\frac{2r}{3}-3r}$

$= {}^{30}C_r (2)^r x^{\frac{60-11r}{3}}$ $\quad 0 \le r \le 30$

For $\beta x^{-\alpha}$, $\alpha > 0$ the smallest value of α is 2 (for r = 6)

$T7 = {}^{30}C_6 (2)^6 x^{-2}$

$\alpha = 2$

22. The remainder on dividing 5^{99} by 11 is _____ . **[JEE(Main)Jan.2023]**

5^{99} को 11 से विभाजित करने पर शेषफल है। **[JEE(Main)Jan.2023]**

Ans. 9

Sol. $5^{99} = 5^4.5^{95}$

$= 625\ [5^5]^{19}$

$= 625\ [3125]^{19}$

$= 625\ [2124 + 1]^{19}$

$= 625\ [11k \times 19 + 1]$

$= 625 \times 11k \times 19 + 625$

$= 11k_1 + 616 + 9$

$= 11(k_2) + 9$

Remainder = 9

23. The remainder, when $19^{200} + 23^{200}$ is divided by 49, is _____. **[JEE(Main)Jan.2023]**

$19^{200} + 23^{200}$ को 49 से विभाजित करने पर शेषफल है। **[JEE(Main)Jan.2023]**

Ans. 29

Sol. $(21 - 2)^{200} + (21 + 2)^{200} = 49\lambda + 2^{201}$

$2^{201} = 8^{67} = (7 + 1)^{67} = 49\lambda + 7 \times 67 + 1$

$= 49\lambda + 470$

$= 49\ (\lambda + 9) + 29$

So, remainder = 29

24. Let the sixth term in the binomial expansion of $\left(\sqrt{2^{\log_2\left(10-3^x\right)}} + \sqrt[5]{2^{(x-2)\log_2 3}}\right)^m$, in the increasing powers of $2^{(x-2)\log_2 3}$, be 21. If the binomial coefficients of the second, third and fourth terms in the expansion are respectively the first, third and fifth terms of an A.P., then the sum of the squares of all possible values of x is _____ . **[JEE(Main)Jan.2023]**

माना $2^{(x-2)\log_2 3}$ की बढ़ती घातों में $\left(\sqrt{2^{\log_2(10-3^x)}}+\sqrt[5]{2^{(x-2)\log_2 3}}\right)^m$ के द्विपद प्रसार में छठा पद 21 है। यदि इस प्रसार में दूसरा, तीसरा तथा चौथा द्विपद गुणांक एक A.P. के क्रमशः पहला, तीसरा तथा पाँचवा पद है, तो x के सभी संभव मानों के वर्गो का योग है। **[JEE(Main)Jan.2023]**

Ans. 4

Sol. $T_6 = {}^mC_5\left(10-3^x\right)^{\frac{m-5}{2}}\cdot\left(3^{x-2}\right)=21$ ...(1)

${}^mC_1, {}^mC_2, {}^mC_3$ are in A.P.

$2.{}^mC_2 = {}^mC_1 + {}^mC_3$

Solving for m, we get

m = 2 (rejected), 7

Put in equation (1)

$$21.\left(10-3^x\right)\frac{3^x}{9}=21$$

$3^x = 3^0, 3^2$

x = 0, 2

Sum of all the squares of all possible values of x = 4

25. If the term without x in the expansion of $\left(x^{\frac{2}{3}}+\frac{\alpha}{x^3}\right)^{22}$ is 7315, then $|\alpha|$ is equal to _____ .

[JEE(Main)Jan.2023]

यदि $\left(x^{\frac{2}{3}}+\frac{\alpha}{x^3}\right)^{22}$ के प्रसार में x से स्वतंत्र पद 7315 है, तो $|\alpha|$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 1

Sol. $T_{r+1} = {}^{22}C_r\cdot\left(x^{\frac{2}{3}}\right)^{22-r}.(\alpha)^r, x^{-3r}$

$$= {}^{22}C_r.x^{\frac{44}{3}-\frac{2r}{3}-3r}(\alpha)^r$$

$$\frac{44}{3}=\frac{11r}{3}$$

r = 4

${}^{22}C_4.ga^4 = 7315$

$$\frac{22\times21\times20\times19}{24}.\alpha^4 = 7315$$

$\alpha = 1$

26. If the ratio of the fifth term from the beginning to the fifth term from the end in the expansion of $\left(\sqrt[4]{2}+\frac{1}{\sqrt[4]{3}}\right)^n$ is $\sqrt{6}:1$, then the third term from the beginning is : **[JEE(Main)April,2023]**

यदि $\left(\sqrt[4]{2}+\frac{1}{\sqrt[4]{3}}\right)^n$ के विस्तार में आरंभ से पाँचवें पद का अंत से पाँचवे पद से अनुपात $\sqrt{6}:1$ है, तब आरंभ से तीसरा पद है : **[JEE(Main)April,2023]**

(A) $30\sqrt{3}$ (B) $60\sqrt{3}$ (C) $60\sqrt{2}$ (D) $30\sqrt{2}$

Ans. B

Sol. $$\frac{{}^nC_4 2^{\frac{n-4}{4}}\cdot\left(3^{\frac{-1}{4}}\right)^4}{{}^nC_4 3^{-\left(\frac{n-4}{4}\right)}\cdot\left(2^{\frac{1}{4}}\right)^4}=\frac{\sqrt{6}}{1}$$

$\Rightarrow n = 10$

So $T_3 = {}^{10}C_2 2^{\frac{1}{4}\cdot 8}\cdot 3^{-\frac{1}{4}\cdot 2} = \frac{45.4}{\sqrt{3}} = 60\sqrt{3}$

27. If the coefficients of x^7 in $\left(ax^2+\frac{1}{2bx}\right)^{11}$ and x^{-7} in $\left(ax-\frac{1}{3bx^2}\right)^{11}$ are equal, then : **[JEE(Main)April,2023]**

यदि $\left(ax^2+\frac{1}{2bx}\right)^{11}$ में x^7 तथा $\left(ax-\frac{1}{3bx^2}\right)^{11}$ में x^{-7} के गुणांक बराबर है, तो : **[JEE(Main)April,2023]**

(A) $64ab = 243$ (B) $243ab = 64$ (C) $729ab = 32$ (D) $32ab = 729$

Ans. C

Sol. Coefficient of x^7 in $\left(ax^2+\frac{1}{2bx}\right)^{11}$

$$T_{r+1} = {}^{11}C_r\left(ax^2\right)^{11-r}\left(\frac{1}{2bx}\right)^r$$

$$= {}^{11}C_r (a)^{11-r}\left(\frac{1}{2b}\right)^r x^{22-3r}$$

$22-3r = 7 \Rightarrow r = 5$

Coefficient of x^{-7} in $\left(ax-\frac{1}{3bx^2}\right)^{11}$

$T_{r+1}={}^{11}C_r(ax)^{11-r}\left(-\frac{1}{3bx^2}\right)^r$

$={}^{11}C_r a^{11-r}\left(-\frac{1}{3b}\right)^r x^{11-3r}$

$\therefore \quad {}^{11-3r}=-7 \Rightarrow r=6$

$\Rightarrow \quad {}^{11}C_5(ab)^6\left(\frac{1}{2b}\right)^5=32$

$\Rightarrow \quad 729ab=32$

28. Among the statements :

(S1) : $2023^{2022}-1999^{2022}$ is divisible by 8

(S2) : $13(13)^n-11n-13$ is divisible by 144 for infinitely many $n\in\mathbb{N}$: **[JEE(Main)April,2023]**

(A) both (S1) and (S2) are correct (B) both (S1) and (S2) are incorrect

(C) only (S2) is correct (D) only (S1) is correct

कथनों :

(S1) : $2023^{2022}-1999^{2022}$, 8 से विभाज्य है

(S2) : $13(13)^n-11n-13$ अनंत $n\in\mathbb{N}$ के लिए 144 से विभाज्य है

में से **[JEE(Main)April,2023]**

(A) दोनों (S1) तथा (S2) सही है (B) दोनों (S1) तथा (S2) गलत है

(C) केवल (S2) सही है (D) केवल (S1) सही है

Ans. D

Sol. Statement – 1 : $x^{2022}-y^{2022}$ is divisible by $(x-y)$

Hence $(2023)^{2022}-(1999)^{2022}$ is divisible by $2023-1999=24$

$\Rightarrow$ statement : 1 is correct

Statement – 2 : $13\,(13)^n-11n-13$

$=13(12+1)^n-11n-13$

$=13\left[12^n+n(12)^{n-1}+\frac{n(n-1)}{2}(12)^{n-2}+\ldots\ldots\ldots\ldots+{}^nC_n\right]-11n-13$

$=13\left[12^n+n(12)^{n-1}+{}^nC_2(12)^{n-2}+\ldots\ldots\ldots+{}^nC_{n-2}(12)^2\right]-11n-13$

$=13\left[12^n+n(12)^{n-1}+{}^nC_2(12)^{n-2}+\ldots\ldots\ldots+{}^nC_{n-2}(12)+1\right]+156n-11n$

$=13\left[12^n+n\cdot12^{n-1}+{}^nC_2(12)^{n-2}+\ldots\ldots\ldots+{}^nC_{n-2}(12)^2\right]+145n$

Which is not divisible by 144 for all $n \in N$ but divisible by 144 for infinitely many values of n if $n = 144m$ where $m \in N$.

29. If the coefficients of three consecutive terms in the expansion of $(1+x)^n$ are in the ratio 1 : 5 : 20, then the coefficient of the fourth term is : **[JEE(Main)April,2023]**

यदि $(1+x)^n$ के प्रसार में तीन क्रमागत पदों के गुणांकों का अनुपात 1 : 5 : 20 है, तो चौथे पद का गुणांक है : **[JEE(Main)April,2023]**

(A) 1827 (B) 2436 (C) 5481 (D) 3654

Ans. D

Sol. $\frac{{}^nC_{r-1}}{1} = \frac{{}^nC_r}{5} = \frac{{}^nC_{r+1}}{20}$

$$\because \quad \frac{{}^nC_r}{{}^nC_{r-1}} = 5 \Rightarrow \frac{\frac{\lfloor n}{\lfloor n-r \rfloor r}}{\frac{\lfloor n}{\lfloor n-r+1 \rfloor r-1}} = 5$$

$$\Rightarrow \frac{n-r+1}{r} = 5 \Rightarrow n = 6r-1 \ldots \text{ (i)}$$

$$\because \quad \frac{{}^nC_{r+1}}{{}^nC_r} = 4 \Rightarrow \frac{n-r}{r+1} = 4 \Rightarrow n = 5r+4.$$

from (i) and (ii), $r = 5, n = 29$

$\Rightarrow$ Coefficient of fourth term $= {}^{29}C_3 = 3654$

30. The absolute difference of the coefficients of x^{10} and x^7 in the expansion of $\left(2x^2 + \frac{1}{2x}\right)^{11}$ is equal to : **[JEE(Main)April,2023]**

$\left(2x^2 + \frac{1}{2x}\right)^{11}$ के प्रसार में x^{10} तथा x^7 के गुणांकों का निरपेक्ष अंतर बराबर है : **[JEE(Main)April,2023]**

(A) $11^3 - 11$ (B) $12^3 - 12$ (C) $13^3 - 13$ (D) $10^3 - 10$

Ans. B

Sol. $T_{r+1} = {}^{11}C_r \left(2x^2\right)^{11-r} (2x)^{-r}$

For coefficient of x^7

$\therefore \quad 22 - 2r - r = 7$

$\therefore \quad r = 5$

For coefficient of x^{10}

$22 - 3r = 10$

$\Rightarrow r = 4$

Absolute difference $= \left| {}^{11}C_5 \cdot \frac{2^6}{2^5} - {}^{11}C_4 \cdot \frac{2^7}{2^4} \right|$

$= |924 - 2640| = 1716$

$= 12^3 - 12$

31. $25^{190} - 19^{190} - 8^{190} + 2^{190}$ is divisible by : **[JEE(Main)April,2023]**

(A) 14 but not by 34 (B) both 14 and 34 (C) 34 but not by 14 (D) neither 14 nor 34

$25^{190} - 19^{190} - 8^{190} + 2^{190}$ **[JEE(Main)April,2023]**

(A) 14 से विभाज्य है परन्तु 34 से नहीं (B) 14 तथा 34 दोनों से विभाज्य है

(C) 34 वे विभाज्य है परन्तु 14 से नहीं (D) न तो 14 से नही 34 से विभाज्य है

Ans. C

Sol. $(25^{190} - 19^{190}) - (8^{190} - 2^{190})$ is divisible by 6.

$(25^{190} - 8^{190}) - (19^{190} - 2^{190})$ is divisible by 17.

$25^{190} - 8^{190}$ is not divisible by 7

but $19^{190} - 2^{190}$ is divisible by 7

So, $25^{190} - 19^{190} - 8^{190} + 2^{190}$ is divisible by 34 but not 14

32. If the coefficient of x^7 in $\left(ax - \frac{1}{bx^2}\right)^{13}$ and the coefficient of x^{-5} in $\left(ax + \frac{1}{bx^2}\right)^{13}$ are equal, then a^4b^4 is equal to : **[JEE(Main)April,2023]**

यदि $\left(ax - \frac{1}{bx^2}\right)^{13}$ में x^7 का गुणांक तथा $\left(ax + \frac{1}{bx^2}\right)^{13}$ में x^{-5} का गुणांक बराबर हैं, तो a^4b^4 बराबर है : **[JEE(Main)April,2023]**

(A) 22 (B) 33 (C) 44 (D) 11

Ans. A

Sol. Coefficient of x^7 in $\left(ax - \frac{1}{bx^2}\right)^{13}$

$T_{r+1} = {}^{13}C_r (9x)^{13-r} \left(-\frac{1}{bx^2}\right)^r$

$\Rightarrow r = 2$

$\therefore$ Coeff. $= {}^{13}C_2 \frac{a^{11}}{b^2}$

Similarly coeff. of x^{-5} in $\left(ax + \frac{1}{bx^2}\right)^{13}$

$\Rightarrow r = 6$

$\therefore$ Coeff. $= {}^{13}C_6 \frac{a^7}{b^6}$

Now, ${}^{13}C_2 \frac{a^{11}}{b^2} = {}^{13}C_6 \frac{a^7}{b^6}$

$\Rightarrow a^4b^4 = 22$

33. Let the number $(22)^{2022} + (2022)^{22}$ leave the remainder α when divided by 3 and β when divided by 7. Then $(\alpha^2 + \beta^2)$ is equal to : **[JEE(Main)April,2023]**

माना $(22)^{2022} + (2022)^{22}$ को 3 से विभाजित करने पर शेषफल α है तथा 7 से विभाजित करने पर शेषफल β है। तो $(\alpha^2 + \beta^2)$ बराबर है : **[JEE(Main)April,2023]**

(A) 13 (B) 5 (C) 20 (D) 10

Ans. B

Sol. For $\alpha : (21+1)^{2022} + \underbrace{(2022)^{22}}_{\text{divisible by 3}}$

$= 3K_1 + 1$

For $\beta : (21+1)^{2022} + (2023-1)^{22}$

$= 7\lambda + 1 + 7\mu + 1$

$= 7K_2 + 2$

So, $\alpha = 1, \beta = 2$

$\alpha^2 + \beta^2 = 5$

34. If the coefficients of x and x^2 in $(1 + x)^p (1 - x)^q$ are 4 and –5 respectively, then $2p + 3q$ is equal to : **[JEE(Main)April,2023]**

$(1 + x)^p (1 - x)^q$ में x तथा x^2 के गुणांक क्रमशः 4 तथा –5 हैं, तो $2p + 3q$ बराबर है : **[JEE(Main)April,2023]**

(A) 63 (B) 69 (C) 60 (D) 66

Ans. A

Sol. $(1+x)^p(1-x)^q$

$= \left(1 + px + \frac{p(p-1)}{2}x^2 + \ldots.\right)\left(1 - qx + \frac{q(q-1)}{2}x^2 + \ldots.\right)$

$= 1 + (p-q)x + \left(\frac{p(p-1)}{2} + \frac{q(q-1)}{2} - pq\right)x^2 + \ldots..$

$p - q = 4 \Rightarrow p = q + 4 \quad \ldots(i)$

$\frac{p(p-1)}{2} + \frac{q(q-1)}{2} - pq = -5$

$\frac{(q+4)(q+3)}{2} + \frac{q(q-1)}{2} - (q+4)q = -5$

$\Rightarrow \quad 6-q=-5$

$\Rightarrow \quad q=11$

$\therefore \quad p=15$

$2p+3q=63$

35. If the 1011th term from the end in the binomial expansion of $\left(\frac{4x}{5}-\frac{5}{2x}\right)^{2022}$ is 1024 times 1011th term from the beginning, then $|x|$ is equal to : **[JEE(Main)April,2023]**

यदि $\left(\frac{4x}{5}-\frac{5}{2x}\right)^{2022}$ के द्विपद प्रसार में अंत से 1011वाँ पद, आरंभ से 1011वें पद का 1024 गुना है, तो $|x|$ बराबर है :

[JEE(Main)April,2023]

(A) 8 (B) 12 (C) 10 (D) 15

Ans. 10 **(Bonus)**

Sol. T_{1011} from beginning $= T_{1010+1}$

$={}^{2022}C_{1010}\left(\frac{4x}{5}\right)^{1012}\left(\frac{-5}{2x}\right)^{1010}$

T_{1011} from end

$={}^{2022}C_{1010}\left(\frac{-5}{2x}\right)^{1012}\left(\frac{4x}{5}\right)^{1010}$

Given : $={}^{2022}C_{1010}\left(\frac{-5}{2x}\right)^{1012}\left(\frac{4x}{5}\right)^{1010}$

$=2^{10}\cdot{}^{2022}C_{1010}\left(\frac{-5}{2x}\right)^{1010}\left(\frac{4x}{5}\right)^{1012}$

$\left(\frac{-5}{2x}\right)^{2}=2^{10}\left(\frac{4x}{5}\right)^{2}$

$x^4=\frac{5^4}{2^{16}}$

$|x|=\frac{5}{16}$

36. The sum of the coefficients of three consecutive terms in the binomial expansion of $(1+x)^{n+2}$, which are in the ratio 1 : 3 : 5, is equal to : **[JEE(Main)April,2023]**

$(1+x)^{n+2}$ के द्विपद प्रसार में तीन क्रमागत पदों के गुणांकों, जो 1 : 3 : 5 के अनुपात में है, का योग बराबर है :

[JEE(Main)April,2023]

(A) 25 (B) 92 (C) 41 (D) 63

Ans. D

Sol. Coefficients are ${}^{n+2}C_{r-1}, {}^{n+2}C_r, {}^{n+2}C_{r+1}$

Now $\frac{{}^{n+2}c_{r-1}}{{}^{n+2}c_r} = \frac{1}{3} \Rightarrow n+3 = 4r$ ....(1)

$\frac{{}^{n+2}c_r}{{}^{n+2}c_{r+1}} = \frac{3}{5} \Rightarrow 3n+1 = 8r$ ....(2)

By(1) & (2) $n = 5, r = 2$

Now coefficients are ${}^{7}C_1, {}^{7}C_2, {}^{7}C_3$

sum = 63

37. If $\frac{1}{n+1}{}^{n}C_n + \frac{1}{n}{}^{n}C_{n-1} + \ldots + \frac{1}{2}{}^{n}C_1 + {}^{n}C_0 = \frac{1023}{10}$ then n is equal to : **[JEE(Main)April,2023]**

यदि $\frac{1}{n+1}{}^{n}C_n + \frac{1}{n}{}^{n}C_{n-1} + \ldots + \frac{1}{2}{}^{n}C_1 + {}^{n}C_0 = \frac{1023}{10}$ तो n बराबर है : **[JEE(Main)April,2023]**

(A) 7 (B) 6 (C) 8 (D) 9

Ans. D

Sol. $\sum_{r=0}^{n} \frac{1}{r+1}{}^{n}C_r = \frac{1023}{10} \quad \left(\because {}^{n+1}C_{r+1} = \frac{n+1}{r+1}{}^{n}C_r\right)$

$\Rightarrow \sum_{r=0}^{n} \frac{1}{n+1}{}^{n+1}C_{r+1} = \frac{1023}{10}$

$\Rightarrow \frac{1}{n+1}\left[{}^{n+1}C_1 + {}^{n+1}C_2 + \ldots + {}^{n+1}C_{n+1}\right] = \frac{1023}{10}$

$\Rightarrow \frac{2^{n+1}-1}{n+1} = \frac{1023}{10} = \frac{2^{10}-1}{10}$

$\Rightarrow n+1 = 10$

$\Rightarrow n = 9$

38. The sum, of the coefficients of the first 50 terms in the binomial expansion of $(1-x)^{100}$, is equal to : **[JEE(Main)April,2023]**

$(1-x)^{100}$ के द्विपद प्रसार में प्रथम 50 पदों के गुणांकों का योग बराबर है : **[JEE(Main)April,2023]**

(A) ${}^{99}C_{49}$ (B) $-{}^{99}C_{49}$ (C) $-{}^{101}C_{50}$ (D) ${}^{101}C_{50}$

Ans. B

Sol. $(1-x)^{100} = Co - C_1x + C_2x^2 - C_3x^3 + \ldots C_{99}x^{99} + C_{100}x^{100}$

$\Rightarrow Co - C_1 + C_2 - C_3 + \ldots\ldots - C_{99} + C_{100} = 0$

$2\left(Co - C_1 + C_2 + \ldots\ldots - C_9\right) + C_{50} = 0$

$$C_0 - C_1 + C_2 + \ldots . C_{99} = -\frac{1}{2}^{100} C_{50}$$

$$-\frac{1}{2}\frac{100!}{50!50!} = -\frac{1}{2} \times \frac{100 \times 99!}{50!50!} = -{}^{99}C_{49}$$

39. Fractional part of the number $\frac{4^{2022}}{15}$ is equal to : **[JEE(Main)April,2023]**

संख्या $\frac{4^{2022}}{15}$ का भिन्न भाग है : **[JEE(Main)April,2023]**

(A) $\frac{4}{15}$ (B) $\frac{14}{15}$ (C) $\frac{1}{15}$ (D) $\frac{8}{15}$

Ans. C

Sol. $\frac{4^{2022}}{15}$

$\because 4 \equiv 4(\text{mod}15)$

$4^2 \equiv 1(\text{mod}15)$

$4^{2022} \equiv 1(\text{mod}15)$

$\therefore$ Fractional part of $\frac{4^{2022}}{15} = \frac{1}{15}$

40. The coefficient of x^5 in the expansion of $\left(2x^3 - \frac{1}{3x^2}\right)^5$ is : **[JEE(Main)April,2023]**

$\left(2x^3 - \frac{1}{3x^2}\right)^5$ के प्रसार में x^5 का गुणांक है : **[JEE(Main)April,2023]**

(A) 9 (B) 8 (C) $\frac{26}{3}$ (D) $\frac{80}{9}$

Ans. D

Sol. General term for $\left(2x^3 - \frac{1}{3x^2}\right)^5$

$T_{r+1} = {}^5C_r\left(-\frac{1}{3x^2}\right)^r\left(2x^3\right)^{5-r}$

$= {}^5C_r(-1)^r 3^{-r} 2^{5-r} \cdot x^{15-5r}$

$15 - 5r = 5 \Rightarrow r = 2$

Coeff. of $x^5 = {}^5C_2(-1)^2 3^{-2} 2^3$

$= 10 \times \frac{1}{9} \times 8$

$= \frac{80}{9}$

41. Let $\left(a + bx + cx^2\right)^{10} = \sum_{i=0}^{20} p_i x^i, a, b, c \in \mathbb{N}$. If $p_1 = 20$ and $p_2 = 210$, then 2(a+b+c) is equal to :

[JEE(Main)April,2023]

यदि $\left(a + bx + cx^2\right)^{10} = \sum_{i=0}^{20} p_i x^i, a, b, c \in \mathbb{N}$, $p_1 = 20$ तथा $p_2 = 210$ हो, तो 2(a+b+c) का मान बराबर होगा :

[JEE(Main)April,2023]

(A) 8 (B) 12 (C) 6 (D) 15

Ans. B

Sol. $\left(a + bx + cx^2\right)^{10} = \sum_{i=0}^{20} p_i x^i,$

Coefficient of $x^1 = 20$

$20 = \frac{10!}{9!1!} \times a^9 \times b^1$

$a^9 \cdot b = 2$

$a = 1, b = 2$

Coefficient of $x^2 = 210$

$210 = \frac{10!}{9!1!} \times a^9 \times c^1 + \frac{10!}{8!2!} \times a^8 b^2$

$210 = 10 \cdot c + 45 \times 4$

$10c = 30$

$c = 3$

$2 (a + b + c) = 12$

42. The coefficient of x^{18} in the expansion of $\left(x^4 - \frac{1}{x^3}\right)^{15}$ is _____. **[JEE(Main)April,2023]**

$\left(x^4 - \frac{1}{x^3}\right)^{15}$ के प्रसार में x^{18} का गुणांक है _____ **[JEE(Main)April,2023]**

Ans. 5005

Sol. $\left(x^4 - \frac{1}{x^3}\right)^{15}$

$T_{r+1} = {}^{15}C_r \left(x^4\right)^{15-r} \left(\frac{-1}{x^3}\right)^r$

$60-7r=18$

$r=6$

Hence coeff. of $x^{18} = {}^{15}C_6 = 5005$

43. Let [t] denote the greatest integer $\le$ t. If the constant term in the expansion of $\left(3x^2-\frac{1}{2x^5}\right)^7$ is α, then $[\alpha]$ is equal to _____. **[JEE(Main)April,2023]**

माना [t] महत्तम पूर्णांक $\le$ t है। यदि $\left(3x^2-\frac{1}{2x^5}\right)^7$ के प्रसार में अचर पद α है, तो $[\alpha]$ बराबर है _____

[JEE(Main)April,2023]

Ans. 1275

Sol. $T_{r+1} = {}^7C_r\left(3x^2\right)^{7-r}\left(-\frac{1}{2x^5}\right)^r$

$T_{r+1} = {}^7C_r 3^{7-r}\left(-\frac{1}{2}\right)^r x^{14-7r}$

For term independent of x : $14-7r=0 \Rightarrow r=2$

Required coefficient $= {}^7C_2 3^{7-2}\left(-\frac{1}{2}\right)^2 = \frac{7.6}{2}\cdot 3^5\cdot\frac{1}{2^2}$

$=\frac{7\cdot 3^6}{4}=1275.85$

[1275.85] = 1275

44. The coefficient of x^7 in $\left(1-x+2x^3\right)^{10}$ is _____. **[JEE(Main)April,2023]**

$\left(1-x+2x^3\right)^{10}$ में x^7 का गुणांक है _____ **[JEE(Main)April,2023]**

Ans. 960

Sol. General term $=\frac{10!}{r_1!\cdot r_2!\cdot r_3!}(-1)^{r_2}\cdot(2)^{r_3}x^{r_2+3r_3}$

where $r_1+r_2+r_3=10$ and $r_2+3r_3=7$

r_1	r_2	r_3
3	7	0
5	4	1
7	1	2

Required coefficient

$=\frac{10!}{3!\cdot 7!}(-1)^7+\frac{10!}{5!\cdot 4!}(-1)^4(2)+\frac{10!}{7!\cdot 2!}(-1)^1(2)^2$

$= -120 + 2520 - 1440 = 960$

45. The number of integral terms in the expansion of $\left(3^{\frac{1}{2}} + 5^{\frac{1}{4}}\right)^{680}$ is equal to _____.

[JEE(Main)April,2023]

$\left(3^{\frac{1}{2}} + 5^{\frac{1}{4}}\right)^{680}$ के प्रसार में पूर्णांक पदों की संख्या है _____ **[JEE(Main)April,2023]**

Ans. 171

Sol. The number of integral term in the expression of $\left(3^{\frac{1}{2}} + 5^{\frac{1}{4}}\right)^{680}$ is equal to

General term $= {}^{680}C_r \left(3^{\frac{1}{2}}\right)^{680-r} \left(5^{\frac{1}{4}}\right)^r$

$= {}^{680}C_r 3^{\frac{680-r}{2}} 5^{\frac{r}{4}}$

Value's of r, where $\frac{r}{4}$ goes to integer

r = 0, 4, 8, 12, ...680

All value of r are accepted for $\frac{680-r}{2}$ as well so

No of integral terms = 171.

46. Let α be the constant term in the binomial expansion of $\left(\sqrt{x} - \frac{6}{x^{\frac{3}{2}}}\right)^n$, $n \le 15$. If the sum of the coefficients of the remaining terms in the expansion is 649 and the coefficient of x^{-n} is $\lambda\alpha$, then λ is equal to _____.

[JEE(Main)April,2023]

माना $\left(\sqrt{x} - \frac{6}{x^{\frac{3}{2}}}\right)^n$, $n \le 15$ के द्विपद प्रसार में अचर पद α है। यदि इस प्रसार में शेष पदों के गुणांकों का योग 649 है तथा x^{-n} का गुणांक $\lambda\alpha$ है, तो λ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 36

Sol. $T_{k+1} = {}^nC_k (x)^{\frac{n-k}{2}} (-6)^k (x)^{\frac{-3}{2}k}$

$\frac{n-k}{2} - \frac{3}{2}k = 0$

$n - 4k = 0$

$$(-5)^n - \left({}^nC_{\frac{n}{4}} (-6)^{\frac{n}{4}} \right) = 649$$

By observation (625 + 24 =649), we get n = 4

$\because$ n = 4 & k = 1

Required is coefficient of x^{-4} is $\left(\sqrt{4} - \frac{6}{x^{\frac{3}{2}}} \right)^4$

${}^4C_1 (-6)^3$

By calculating we will get $\lambda = 36$

47. The remainder, when 7^{103} is divided by 17, is _____. **[JEE(Main)April,2023]**

7^{103} को 17 से विभाजित करने पर शेषफल है _____ **[JEE(Main)April,2023]**

Ans. 12

Sol. 7^{103} divided by 17

$7 \equiv 7 \pmod{17}$

$7^2 \equiv -2 \pmod{17}$

$7^6 \equiv -8 \pmod{17}$

$7^8 \equiv -1 \pmod{17}$

$7^{16} \equiv 1 \pmod{17}$

$7^{103} \equiv 12 \pmod{17}$

$\therefore$ Remainder = 12

48. The number of elements in the set $\{n \in \mathbb{N} : 10 \le n \le 100$ and $3^n - 3$ is a multiple of $7\}$ is _____.
[JEE(Main)April,2023]

समुच्चय में अवयवों की संख्या होगी $\{n \in \mathbb{N} : 10 \le n \le 100$ तथा $3^n - 3$, 7 का गुणक $\}$_____
[JEE(Main)April,2023]

Ans. 15

Sol. $n \in [10, 100]$

$3^n - 3$ is multiple of 7

$3^n = 7\lambda + 3$

n = 1,7,13,20,.....97

Number of possible values of n = 15

49. If A denotes the sum of all the coefficients in the expansion of $(1 - 3x + 10x^2)^n$ and B denotes the sum of all the coefficients in the expansion of $(1 + x^2)^n$, then : **[JEE(Main)Jan.2024]**

यदि $(1 - 3x + 10x^2)^n$ के प्रसार में सभी गुणांकों का योग A है तथा $(1 + x^2)^n$ के प्रसार में सभी गुणांकों का योग B है, तो :
[JEE(Main)Jan.2024]

(A) $B = A^3$ (B) $3A = B$ (C) $A = B^3$ (D) $A = 3B$

Ans. C

Sol. $A = \text{sum of coefficient} = (1 - 3(1) + 10(1)^2)^n = 8^n$

(Putting $x = 1$)

$B = \text{Sum of coefficient} = (1 + (1)^2)^n = 2^n$

$\Rightarrow 8^n = (23)^n = (2^n)^3 = B^3$

$\Rightarrow A = B^3$

50. ${}^{n-1}C_r = (k^2 - 8)\,{}^nC_{r+1}$ if and only if : **[JEE(Main)Jan.2024]**

${}^{n-1}C_r = (k^2 - 8)\,{}^nC_{r+1}$ है यदि और केवल यदि : **[JEE(Main)Jan.2024]**

(A) $2\sqrt{2} < k \le 3$ (B) $2\sqrt{3} < k < 3\sqrt{3}$

(C) $2\sqrt{3} < k \le 3\sqrt{2}$ (D) $2\sqrt{2} < k < 2\sqrt{3}$

Ans. A

Sol. ${}^{n-1}C_r = \left(k^2 - 8\right)\frac{n}{r+1}\,{}^{n-1}C_r$

$k^2 - 8 = \frac{r+1}{n}$

$0 < k^2 - 8 \le 1$

$8 < k^2 \le 9$

$2\sqrt{2} < k \le 3$ or $-3 \le k < -2\sqrt{2}$

51. Suppose $2 - p, p, 2 - \alpha, \alpha$ are the coefficients of four consecutive terms in the expansion of $(1 + x)^n$. Then the value of $p^2 - \alpha^2 + 6\alpha + 2p$ equals : **[JEE(Main)Jan.2024]**

माना $(1 + x)^n$ के प्रसार में चार क्रमागत पदों के गुणांक $2 - p, p, 2 - \alpha, \alpha$ हैं। तो $p^2 - \alpha^2 + 6\alpha + 2p$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) 8 (B) 10 (C) 6 (D) 4

Ans. Bonus

Sol. $2 - p, p, 2 - \alpha, \alpha$

Binomial coefficients are

${}^nC_r, {}^nC_{r+1}, {}^nC_{r+2}, {}^nC_{r+3}$ respectively

$\Rightarrow {}^nC_r + {}^nC_{r+1} = 2$

$\Rightarrow {}^{n+1}C_{r+1} = 2$ _____(1)

Also, ${}^nC_{r+2} + {}^nC_{r+3} = 2$

$\Rightarrow {}^{n+1}C_{r+3} = 2$ _____(2)

From (1) and (2)

${}^{n+1}C_{r+1} = {}^{n+1}C_{r+3}$

$\Rightarrow 2r + 4 = n + 1$

$n = 2r + 3$

${}^{2r+4}C_{r+1} = 2$

Data Inconsistent

52. Let m and n be the coefficients of seventh and thirteenth terms respectively in the expansion of $\left(\frac{1}{3}x^{\frac{1}{3}} + \frac{1}{2x^{\frac{2}{3}}}\right)^{18}$.

Then $\left(\frac{n}{m}\right)^{\frac{1}{3}}$ is : **[JEE(Main)Jan.2024]**

माना $\left(\frac{1}{3}x^{\frac{1}{3}} + \frac{1}{2x^{\frac{2}{3}}}\right)^{18}$ के प्रसार में सातवें तथा तेरहवें पदों के गुणांक क्रमशः m तथा n है। तो $\left(\frac{n}{m}\right)^{\frac{1}{3}}$ बराबर है :

[JEE(Main)Jan.2024]

(A) $\frac{4}{9}$ (B) $\frac{1}{9}$ (C) $\frac{1}{4}$ (D) $\frac{9}{4}$

Ans. D

Sol. $m = {}^{18}C_6\left(\frac{1}{3}\right)^{12}\left(\frac{1}{2}\right)^6$

$n = {}^{18}C_{12}\left(\frac{1}{3}\right)^6\left(\frac{1}{2}\right)^{12}$

$\Rightarrow \frac{n}{m} = \frac{3^6}{2^6}$

$\Rightarrow \left(\frac{n}{m}\right)^{\frac{1}{3}} = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$

53. The coefficient of x^{2012} in the expansion of $(1-x)^{2008}(1+x+x^2)^{2007}$ is equal to _____.

[JEE(Main)Jan.2024]

$(1 - x)^{2008}(1 + x + x^2)^{2007}$ के प्रसार में x^{2012} का गुणांक बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 0

Sol. $\left((1-x)^{2008}\right)\left[1+x+x^2\right]^{2007}$

$(1-x)\left[1-x^3\right]^{2007}$

$(1-x)^{2007}c_r\left(-x^3\right)^r$

$^{2007}c_r\left[(-x)^{3r}+(-x)^{3r+1}\right]$

$\therefore$ 3r = 2012

or 3r + 1 = 2012

no such r exists

$\therefore$ Coefficient is zero

54. If $\frac{^{11}C_1}{2}+\frac{^{11}C_2}{3}+...+\frac{^{11}C_9}{10}=\frac{n}{m}$ with gcd(n, m) = 1, then n + m is equal to _____.

[JEE(Main)Jan.2024]

यदि $\frac{^{11}C_1}{2}+\frac{^{11}C_2}{3}+...+\frac{^{11}C_9}{10}=\frac{n}{m}$ है तथा gcd(n, m) = 1 है, तो n + m बराबर है _____

[JEE(Main)Jan.2024]

Ans. 2041

Sol. $\sum_{r=1}^{9}\frac{^{11}C_r}{r+1}$

$=\frac{1}{12}\sum_{r=1}^{9}{}^{12}C_{r+1}$

$=\frac{1}{12}\left[2^{12}-26\right]=\frac{2035}{6}$

$\therefore$ m + n = 2041

55. Remainder when $64^{32^{32}}$ is divided by 9 is equal to _____ . **[JEE(Main)Jan.2024]**

$64^{32^{32}}$ को 9 से विभाजित करने पर शेषफल है _____ . **[JEE(Main)Jan.2024]**

Ans. 1

Sol. Let $32^{32} = t$

$64^{32^{32}} = 64^t = 8^{2t} = (9-1)^{2t}$

$= 9k + 1$

Hence remainder = 1

56. Number of the integral terms in the expansion of $\left\{7^{\left(\frac{1}{2}\right)}+11^{\left(\frac{1}{6}\right)}\right\}^{824}$ is equal to _____.

[JEE(Main)Jan.2024]

$\left\{7^{\left(\frac{1}{2}\right)}+11^{\left(\frac{1}{6}\right)}\right\}^{824}$ के प्रसार में पूर्णांक पदों की संख्या है _____। **[JEE(Main)Jan.2024]**

Ans. 138

Sol. $T_{r+1} = {}^{824}C_r (7)^{\frac{824-r}{2}} (11)^{\frac{r}{6}}$

$\Rightarrow {}^{824}C_r (7)^{412-\frac{r}{2}} (11)^{\frac{r}{6}}$

$r = 0,6,12,.....,822$

Total 138

57. Let $\alpha = \sum_{k=0}^{n} \left(\frac{\left({}^{n}C_k\right)^2}{k+1} \right)$ and $\beta = \sum_{k=0}^{n-1} \left(\frac{{}^{n}C_k \, {}^{n}C_{k+1}}{k+2} \right)$. If $5\alpha = 6\beta$, then n equals _____.

[JEE(Main)Jan.2024]

माना $\alpha = \sum_{k=0}^{n} \left(\frac{\left({}^{n}C_k\right)^2}{k+1} \right)$ तथा $\beta = \sum_{k=0}^{n-1} \left(\frac{{}^{n}C_k \, {}^{n}C_{k+1}}{k+2} \right)$ हैं। यदि $5\alpha = 6\beta$ हैं, तो n बराबर है _____

[JEE(Main)Jan.2024]

Ans. 10

Sol. $\alpha = \sum_{k=0}^{n} \frac{{}^{n}C_k \cdot {}^{n}C_k}{k+1} \cdot \frac{n+1}{n+1}$

$= \frac{1}{n+1} \sum_{k=0}^{n} {}^{n+1}C_{k+1} \cdot {}^{n}C_{n-k}$

$\alpha = \frac{1}{n+1} \cdot {}^{2n+1}C_{n+1}$

$\beta = \sum_{k=0}^{n-1} {}^{n}C_k \cdot \frac{{}^{n}C_{k+1}}{k+2} \frac{n+1}{n+1}$

$\frac{1}{n+1} \sum_{k=0}^{n-1} {}^{n}C_{n-k} \cdot {}^{n+1}C_{k+2}$

$= \frac{1}{n+1} \cdot {}^{2n+1}C_{n+2}$

$\frac{\beta}{\alpha} = \frac{{}^{2n+1}C_{n+2}}{{}^{2n+1}C_{n+1}} = \frac{2n+1-(n+2)+1}{n+2}$

$\frac{\beta}{\alpha} = \frac{n}{n+2} = \frac{5}{6}$

n = 10

58. In the expansion of $(1+x)\left(1-x^2\right)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5$, $x \neq 0$, the sum of the coefficients of x^3 and x^{-13} is equal to _____ . **[JEE(Main)Jan.2024]**

$(1+x)\left(1-x^2\right)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5$, $x \neq 0$, के प्रसार में x^3 तथा x^{-13} के गुणांकों का योग है _____

[JEE(Main)Jan.2024]

Ans. 118

Sol. $(1+x)\left(1-x^2\right)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5$

$=(1+x)\left(1-x^2\right)\left(\left(1+\frac{1}{x}\right)^3\right)^5$

$=\frac{(1+x)^2(1-x)(1+x)^{15}}{x^{15}}$

$=\frac{(1+x)^{17}-x(1+x)^{17}}{x^{15}}$

= coeff (x^3) in the expansion = coeff(x^{18}) in $(1+x)^{17}-x(1+x)^{17}$

$= 0 - 1$

$= -1$

coeff (x^{-13}) in the expansion = coeff (x^2) in $(1+x)^{17}-x(1+x)^{17}$

$=\binom{17}{2}-\binom{17}{1}$

$= 17 \times 8 - 17$

$= 17 \times 7$

$= 119$

Hence Answer = 119 – 1 = 118

59. Let the coefficient of x^r in the expansion of $(x+3)^{n-1}+(x+3)^{n-2}(x+2)+(x+3)^{n-3}(x+2)^2+.....+(x+2)^{n-1}$ be α_r. If $\sum_{r=0}^{n}\alpha_r=\beta^n-\gamma^n, \beta,\gamma \in \mathbb{N}$, then the value of $\beta^2+\gamma^2$ equals _____. **[JEE(Main)Jan.2024]**

माना $(x+3)^{n-1}+(x+3)^{n-2}(x+2)+(x+3)^{n-3}(x+2)^2+.....+(x+2)^{n-1}$ के प्रसार में x^r का गुणांक α_r है। यदि

$\sum_{r=0}^{n}\alpha_r = \beta^n - \gamma^n, \beta, \gamma \in \mathbb{N}$ है, तो $\beta^2 + \gamma^2$ बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 25

Sol. $(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + \ldots.(x+2)^{n-1}$

$$= (x+3)^{n-1}\frac{\left\{1-\left(\frac{x+2}{x+3}\right)^n\right\}}{1-\left(\frac{x+2}{x+3}\right)}$$

$$= (x+3)^n - (x+2)^n$$

$$\sum_{r=0}^{n}\alpha_r = \sum_{r=0}^{n}\left[(x+3)^n - (x+2)^n\right] = \beta^n - \gamma^n$$

$$\Rightarrow 4^n - 3^n = \beta^n - \gamma^n$$

$$\Rightarrow \beta = 4, \quad \gamma = 3$$

$$\beta^2 + \gamma^2 = 25$$

60. If the coefficient of x^{30} in the expansion $\left(1+\frac{1}{x}\right)^6\left(1+x^2\right)^7\left(1-x^3\right)^8$; $x \neq 0$ is α, then $|\alpha|$ equals _____.

[JEE(Main)Jan.2024]

यदि $\left(1+\frac{1}{x}\right)^6\left(1+x^2\right)^7\left(1-x^3\right)^8$; $x \neq 0$, के प्रसार में x^{30} का गुणांक α है, तो $|\alpha|$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 678

Sol. coefficient of x^{30} in $\frac{(x+1)^6\left(1+x^2\right)^7\left(1-x^3\right)^8}{x^6}$

coefficient of x^{36} in $(1+x)^6\left(1+x^2\right)^7\left(1-x^3\right)^8$

General term

${}^6C_{r_1}\,{}^7C_{r_2}\,{}^8C_{r_3}(-1)^{r_1}\,x^{r_1+2r_2+3r_3}$

$r_1 + 2r_2 + 3r_3 = 36$

Case – I :

r_1	r_2	r_3
0	6	8
2	5	8
4	4	8
4	3	8

$r_1 + 2r_2 = 12$ (taking $r_3 = 8$)

Case – II :

r_1	r_2	r_3
1	7	7
3	6	7
5	5	7

$r_1 + 2r_2 = 15$ (taking $r_3 = 7$)

Case – III :

r_1	r_2	r_3
4	7	6
6	6	6

$r_1 + 2r_2 = 18$ (taking $r_3 = 6$)

Coefficient = $7 + (15 \times 21) + (15 \times 35) + (35)$
$-(6 \times 8) - (20 \times 7 \times 8) - (6 \times 21 \times 8) + (15 \times 28)$
$+ (7 \times 28) = -678 = \alpha$
$|\alpha| = 678$

61. The sum of all rational terms in the expansion of $\left(2^{\frac{1}{5}} + 5^{\frac{1}{3}}\right)^{15}$ is equal to : **[JEE(Main)April,2024]**

$\left(2^{\frac{1}{5}} + 5^{\frac{1}{3}}\right)^{15}$ के प्रसार में सभी परिमेय पदों का योग है : **[JEE(Main)April,2024]**

(A) 6131 (B) 3133 (C) 931 (D) 633

Ans. B

Sol. $T_{r+1} = {}^{15}C_r \left(5^{\frac{1}{3}}\right)^r \left(2^{\frac{1}{5}}\right)^{15-r}$

$= {}^{15}C_r 5^{\frac{r}{3}} \cdot 2^{\frac{15-r}{5}}$

$R = 3\lambda, 15\mu$

$\Rightarrow r = 0, 15$

2 rational terms

$\Rightarrow {}^{15}C_0 2^3 + {}^{15}C_{15}(5)^5$

$= 8 + 3125 = 3133$

62. If the coefficients of x^4, x^5 and x^6 in the expansion of $(1 + x)^n$ are in the arithmetic progression, then the maximum value of n is : **[JEE(Main)April,2024]**

यदि $(1 + x)^n$ के प्रसार में x^4, x^5 तथा x^6 के गुणांक समांतर श्रेणी में हैं, तो n का उच्चतम मान है :

[JEE(Main)April,2024]

(A) 28 (B) 7 (C) 21 (D) 14

Ans. D

Sol. $2^nC_5 = {}^nC_4 + {}^nC_6$

$$\frac{2\cdot n!}{5!(n-5)!} = \frac{n!}{4!(n-4)!} + \frac{n!}{6!(n-6)!}$$

$$\frac{2!}{5\times 4!(n-5)(n-6)!} = \frac{1}{4!(n-4)(n-5)(n-6)!} + \frac{1}{6\times 5\times 4!(n-6)!}$$

$$\frac{2}{5(n-5)} = \frac{1}{(n-4)(n-5)} + \frac{1}{30}$$

$$\frac{2(n-4)}{5} = 1 + \frac{(n-4)(n-5)}{30}$$

$12n - 40 = 30 + n^2 - 9n + 20$

$n^2 - 21n + 98 = 0$

$n = 14, 7$

63. If the constant term in the expansion of $\left(\frac{\sqrt[5]{3}}{x} + \frac{2x}{\sqrt[3]{5}}\right)^{12}$, $x \neq 0$ is $\alpha \times 2^8 \times \sqrt[5]{3}$, then 25α is equal to : **[JEE(Main)April,2024]**

यदि $\left(\frac{\sqrt[5]{3}}{x} + \frac{2x}{\sqrt[3]{5}}\right)^{12}$, $x \neq 0$ के प्रसार में अचर पद $\alpha \times 2^8 \times \sqrt[5]{3}$ है, तो 25α बराबर है : **[JEE(Main)April,2024]**

(A) 724 (B) 742 (C) 693 (D) 639

Ans. C

Sol. $${}^{12}C_r \left(\frac{\sqrt[5]{3}}{x}\right)^{12-r} \left(\frac{2x}{\sqrt[3]{5}}\right)^r = {}^{12}C_r \frac{3^{\frac{12-r}{5}}.2^r}{5^{r/3}} \times x^{2r-12}$$

$2r - 12 = 0 \Rightarrow r = 6$

$$\alpha \times 2^8 \times \sqrt[5]{3} = {}^{12}C_6 \frac{3^{\frac{6}{5}}2^6}{5^2} = \frac{11.3.4.7.3.3^{\frac{1}{5}}2^6}{5^2}$$

$$\alpha = \frac{3^2.7.11}{5^2} \Rightarrow 25\alpha = 693$$

64. Let $0 \le r \le n$. If ${}^{n+1}C_{r+1} : {}^{n}C_{r} : {}^{n-1}C_{r-1} = 55 : 35 : 21$, then $2n + 5r$ is equal to : **[JEE(Main)April,2024]**

माना $0 \le r \le n$ है। यदि ${}^{n+1}C_{r+1} : {}^{n}C_{r} : {}^{n-1}C_{r-1} = 55 : 35 : 21$ है, तो $2n + 5r$ बराबर है : **[JEE(Main)April,2024]**

(A) 55 (B) 60 (C) 62 (D) 50

Ans. D

Sol. $\frac{n+1}{r+1} \times {}^{n}C_{r} : {}^{n}C_{r} : \frac{r}{n} {}^{n}C_{r} = 55 : 35 : 21$

$\Rightarrow \frac{n+1}{r+1} = \frac{55}{34}$ and $\frac{n}{r} = \frac{35}{21}$

$\Rightarrow 7n - 11r = 4$ .....(1)

$3n - 5r = 4$ .....(2)

Solving (1) and (2)

$r = 6$ and $n = 10$

$\Rightarrow 2n + 5r = 20 + 30 = 50$

65. If the term independent of x in the expansion of $\left(\sqrt{a}x^2 + \frac{1}{2x^3}\right)^{10}$ is 105, then a^2 is equal to : **[JEE(Main)April,2024]**

यदि $\left(\sqrt{a}x^2 + \frac{1}{2x^3}\right)^{10}$ के प्रसार में x से स्वतंत्र पद 105 है, तो a^2 बराबर है : **[JEE(Main)April,2024]**

(A) 4 (B) 2 (C) 9 (D) 6

Ans. A

Sol. $T_{r+1} = {}^{10}C_r \left(\sqrt{a}x^2\right)^{10-r} \left(\frac{1}{2x^3}\right)^r$

$2(10 - r) - 3r = 0$

$r = 4$

$T_5 = {}^{10}C_4 \left(\sqrt{a}\right)^6 \left(\frac{1}{2}\right)^4 = 105$

$$= \frac{10.9.8.7}{4.3.2.1} \times \frac{a^3}{2.2.2.2} = 5 \times 7 \times 3z$$

$a^3 = 8$, $a = 2 \Rightarrow a^2 = 4$

66. The coefficient of x^{70} in $x^2(1+x)^{98} + x^3(1+x)^{97} + x^4(1+x)^{96} + ... + x^{54}(1+x)^{46}$ is ${}^{99}C_p - {}^{46}C_q$.. Then a possible value of $p + q$ is : **[JEE(Main)April,2024]**

$x^2(1+x)^{98} + x^3(1+x)^{97} + x^4(1+x)^{96} + ... + x^{54}(1+x)^{46}$ में x^{70} ${}^{99}C_p - {}^{46}C_q$ है। तो $p + q$ का एक संभव मान है : **[JEE(Main)April,2024]**

(A) 68 (B) 83 (C) 55 (D) 61

Ans. B

Sol. $x^2(1+x)^{98} + x^3(1+x^{97}) + x^4(1+x)^{96} + ...x^{54}(1+x)^{46}$

Coeff. of x^{70} : ${}^{98}C_{68}$: + ${}^{97}C_{67}$ + ${}^{96}C_{66}$ + ... ${}^{47}C_{17}$ + ${}^{46}C_{16}$

$= {}^{46}C_{30} + {}^{47}C_{30} + ...{}^{98}C_{30}$

$= ({}^{46}C_{31} + {}^{46}C_{30}) + {}^{47}C_{30} + ... {}^{98}C_{30} - {}^{46}C_{31}$

$= {}^{47}C_{31} + {}^{47}C_{30} + ...{}^{98}C_{30} - {}^{46}C_{31}$ .....

$= {}^{99}C_{31} - {}^{46}C_{31} = {}^{99}C_p - {}^{46}C_q$

Possible values of $(p + q)$ are 62, 83, 99, 46

$\Rightarrow p + q = 83$

67. The sum of the coefficient of $x^{\frac{2}{3}}$ and $x^{-\frac{2}{5}}$ in the binomial expansion $\left(x^{\frac{2}{3}} + \frac{1}{2}x^{-\frac{2}{5}}\right)^9$ is : **[JEE(Main)April,2024]**

$\left(x^{\frac{2}{3}} + \frac{1}{2}x^{-\frac{2}{5}}\right)^9$ के द्विपद प्रसार में $x^{\frac{2}{3}}$ तथा $x^{-\frac{2}{5}}$ के गुणांकों का योग है : **[JEE(Main)April,2024]**

(A) $\frac{63}{16}$ (B) $\frac{21}{4}$ (C) $\frac{19}{4}$ (D) $\frac{69}{16}$

Ans. B

Sol. $T_{r+1} = {}^9C_r\left(x^{2/3}\right)^{9-r} \cdot \left(\frac{1}{2}x^{-\frac{2}{5}}\right)^r = {}^9C_r\left(\frac{1}{2}\right)^r \cdot x^{6-\frac{2r}{3}-\frac{2r}{5}}$

$= {}^9C_r \cdot \frac{1}{2^r} \cdot x^{6-\frac{16r}{15}}$

(I) $6-\frac{16r}{15}=\frac{2}{3}\Rightarrow r=5$ and (II) $6-\frac{16r}{15}=\frac{-2}{3}\Rightarrow 6+\frac{2}{5}=\frac{16r}{15}=\frac{32}{5}\quad\Rightarrow r=6$

$\text{Sum}={}^{9}C_5\cdot\frac{1}{2^5}+{}^{9}C_6\cdot\frac{1}{2^6}=\frac{1}{2^6}\left\{2\cdot{}^{9}C_4+{}^{9}C_3\right\}=\frac{1}{2^6}\left({}^{9}C_4+{}^{10}C_4\right)$

$=\frac{21}{4}$

68. Let $a=1+\frac{{}^{2}C_2}{3!}+\frac{{}^{3}C_2}{4!}+\frac{{}^{4}C_2}{5!}+.....,$

$b=1+\frac{{}^{1}C_0+{}^{1}C_1}{1!}+\frac{{}^{2}C_0+{}^{2}C_1+{}^{2}C_2}{2!}+\frac{{}^{3}C_0+{}^{3}C_1+{}^{3}C_2+{}^{3}C_3}{3!}+.....$

Then $\frac{2b}{a^2}$ is equal to _____. **[JEE(Main)April,2024]**

माना $a=1+\frac{{}^{2}C_2}{3!}+\frac{{}^{3}C_2}{4!}+\frac{{}^{4}C_2}{5!}+.....,$

$b=1+\frac{{}^{1}C_0+{}^{1}C_1}{1!}+\frac{{}^{2}C_0+{}^{2}C_1+{}^{2}C_2}{2!}+\frac{{}^{3}C_0+{}^{3}C_1+{}^{3}C_2+{}^{3}C_3}{3!}+.....$

हैं, तो $\frac{2b}{a^2}$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 8

Sol. $a=1+\frac{2c_2}{\lfloor 3}+\frac{3c_2}{\lfloor 4}+\frac{4c_2}{\lfloor 5}+\cdots$

$a=1+\sum_{r=2}^{\infty}\frac{{}^{r}C_2}{(r+1)}$

$a=1+\sum_{r=2}^{\infty}\frac{r(r-1)}{2\lfloor r+1}$

$a=1+\frac{1}{2}\sum_{r=2}^{\infty}\frac{r(r+1)-2r}{\lfloor (r+1)}$

$a=1+\frac{1}{2}\left[\sum_{r=2}^{\infty}\frac{1}{\lfloor r-1}-\frac{2}{\lfloor r}+\frac{2}{\lfloor r+1}\right]$

$a=1+\frac{1}{2}\left[(e-1)-2(e-2)+2\left(e-2-\frac{1}{2}\right)\right]$

$$a = 1 + \frac{1}{2}[e-2] = \frac{e}{2}$$

$$b = 1 + \frac{{}^1C_0 + {}^1C_1}{1!} + \frac{{}^2C_0 + {}^2C_1 + {}^2C_2}{2!} + \frac{{}^3C_0 + {}^3C_1 + {}^3C_2 + {}^3C_3}{3!} + \ldots$$

$b = e^2$

Now, $= \frac{2b}{a^2} = \frac{2 \times e^2}{e^2} \times 4 = 8$

69. If the constant term in the expansion of $\left(1 + 2x - 3x^3\right)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ is p, then 108p is equal to _____.

[JEE(Main)April,2024]

यदि $\left(1 + 2x - 3x^3\right)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ के प्रसार में अचर पद p है, तो 108p बराबर है _____

[JEE(Main)April,2024]

Ans. 54

Sol. General term of $\left(\frac{3}{2} \times^2 - \frac{1}{3x}\right)^9$

$$T_{r+1} = {}^9C_r \left(\frac{3}{2}x^2\right)^{9-r} \left(-\frac{1}{3x}\right)^r = {}^9C_r (-1)^r 3^{9-2r} 2^{r-9} x^{18-3r}$$

Constant term in expansion of $(1 + 2x - 3x^3)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$

$= T_7 - 3T_8 = {}^9C_6\, 3^{-3}.\, 2^{-3} + 3{}^9C_7\, .\, 3^{-5}\, .\, 2^{-2}$

$$= \frac{3 \times 4 \times 7}{3^3.2^3} + \frac{3 \times 9 \times 4}{3^5 \times 2^2} =$$

$$p = \frac{42 + 12}{108} = \frac{54}{108}$$

$108p = 54$

70. If $1 + \frac{\sqrt{3} - \sqrt{2}}{2\sqrt{3}} + \frac{5 - 2\sqrt{6}}{18} + \frac{9\sqrt{3} - 11\sqrt{2}}{36\sqrt{3}} + \frac{49 - 20\sqrt{6}}{180}$ +.....upto $\infty = 2 + \left(\sqrt{\frac{b}{a}} + 1\right)\log_e\left(\frac{a}{b}\right)$, where a and b are integers with gcd (a, b) = 1, then 11a + 18b is equal to _____. **[JEE(Main)April,2024]**

यदि $1+\frac{\sqrt{3}-\sqrt{2}}{2\sqrt{3}}+\frac{5-2\sqrt{6}}{18}+\frac{9\sqrt{3}-11\sqrt{2}}{36\sqrt{3}}+\frac{49-20\sqrt{6}}{180}+.....\text{upto}\infty=2+\left(\sqrt{\frac{b}{a}}+1\right)\log_e\left(\frac{a}{b}\right)$, है, जहाँ a तथा b पूर्णांक हैं तथा gcd(a, b) = 1 है, तो 11a + 18b बराबर है _____ **[JEE(Main)April,2024]**

Ans. 76

Sol. $S = 1 + \frac{\sqrt{3}-\sqrt{2}}{2\sqrt{3}}+\frac{5-2\sqrt{6}}{18}+\frac{9\sqrt{3}-11\sqrt{2}}{36\sqrt{3}}+.....\infty$

$= 1 + \frac{\left(1-\sqrt{2}/\sqrt{3}\right)}{2}+\frac{\left(1-\sqrt{2}/\sqrt{3}\right)^2}{6}+\frac{\left(1-\sqrt{2}/\sqrt{3}\right)^3}{12}+.....\infty$

let $1-\frac{\sqrt{2}}{\sqrt{3}}=a$

$S = 1 + \frac{a}{2}+\frac{a^2}{6}+\frac{a^3}{12}+.....$

$=1+\left(1-\frac{1}{2}\right)a+\left(\frac{1}{2}-\frac{1}{3}\right)a^2+\left(\frac{1}{3}-\frac{1}{4}\right)a^3+....$

$=1+\left(a+\frac{a^2}{2}+\frac{a^3}{3}....\infty\right)+\frac{1}{a}\left(\frac{-a^2}{2}-\frac{a^3}{3}-\frac{a^4}{4}....\infty\right)$

$=-\text{In}(1-a)+\frac{1}{a}\left(-a-\frac{a^2}{2}-\frac{a^3}{3}....\infty\right)+2$

$=-\text{In}(1-a)+\frac{1}{a}\text{In}(1-a)+2$

$=2+\left(\frac{1}{a}-1\right)\text{In}(1-a)$

$=2+\left(\frac{\sqrt{3}}{\sqrt{3-\sqrt{2}}}-1\right)\text{In}\left(1-1+\sqrt{\frac{2}{3}}\right)$

$=2+\frac{\sqrt{2}}{\sqrt{3}-\sqrt{2}}\ \text{In}\sqrt{\frac{2}{3}}$

$=2+\left(\frac{\sqrt{6}+2}{1}.\frac{1}{2}\text{In}\frac{2}{3}\right)$

$= 2 + \left(\sqrt{\frac{3}{2}} + 1\right) \ln \frac{2}{3}$

$\therefore 11a + 18b = 76$

71. If the second, third and fourth terms in the expansion of $(x + y)^n$ are 135, 30 and $\frac{10}{3}$, respectively, then $6\left(n^3 + x^2 + y\right)$ is equal to _____. **[JEE(Main)April,2024]**

यदि $(x + y)^n$ के प्रसार में द्वितीय, तृतीय व चतुर्थ पद क्रमशः 135, 30 तथा $\frac{10}{3}$ हैं, तो $6\left(n^3 + x^2 + y\right)$ बराबर है _____

[JEE(Main)April,2024]

Ans. 806

Sol. ${}^nC_1 x^{n-1} y = 135$ _____(i)

${}^nC_2 x^{n-2} y^2 = 30$ _____(ii)

${}^nC_3 x^{n-3} y^3 = \frac{10}{3}$ _____(iii)

By $\frac{(i)}{(ii)}$

$\frac{{}^nC_1}{{}^nC_2} \frac{x}{y} = \frac{9}{2}$ _____(iv)

By $\frac{(ii)}{(iii)}$

$\frac{{}^nC_2}{{}^nC_3} \frac{x}{y} = 9$ _____(v)

By $\frac{(iv)}{(v)}$

$\frac{{}^nC_1 {}^nC_3}{{}^nC_2 {}^nC_2} = \frac{1}{2}$

$\frac{2n^2(n-1)(n-2)}{6} = \frac{n(n-1)}{2} \frac{n(n-1)}{2}$

$4n - 8 = 3n - 3$

$\Rightarrow n = 5$

put in (v)

$\frac{x}{y} = 9$

$x = 9y$

put in (i)

$^5C_1 x^4 \left(\frac{x}{9}\right) = 135$

$x^5 = 27 \times 9$

$\Rightarrow x = 3, \qquad y = \frac{1}{3}$

$6(n^3 + x^2 + y)$

$= 6\left(125 + 9 + \frac{1}{3}\right)$

$= 806$

72. Let $\alpha = \sum_{r=0}^{n}\left(4r^2 + 2r + 1\right)\ {}^nC_r$ and $\beta = \left(\sum_{r=0}^{n}\frac{{}^nC_r}{r+1}\right) + \frac{1}{n+1}$. If $140 < \frac{2\alpha}{\beta} < 281$, then the value of n is _____.

[JEE(Main)April,2024]

माना $\alpha = \sum_{r=0}^{n}\left(4r^2 + 2r + 1\right)\ {}^nC_r$ तथा $\beta = \left(\sum_{r=0}^{n}\frac{{}^nC_r}{r+1}\right) + \frac{1}{n+1}$ है। यदि $140 < \frac{2\alpha}{\beta} < 281$, तब n का मान होगा _____

[JEE(Main)April,2024]

Ans. 5

Sol. $\alpha = \sum_{r=0}^{n} r^2\ {}^nC_r + 2\sum_{r=0}^{n} r\ {}^nC_r + {}^nC_r$

$\sum (4 \cdot r(r-1) + 6r + 1)\,{}^nC_r$

$4.n(n-1)2^{n-2} + 6 \cdot n \cdot 2^{n-1} + 2^n$

$= 2^n (n+1)^2$

$\beta = \sum_{r=0}^{n}\frac{{}^nC_r}{r+1} + \frac{1}{n+1} = \frac{1}{n+1}\cdot\left[2^{n+1}-1\right] + \frac{1}{n+1} = \frac{2^{n+1}}{n+1}$

Now, $140 < \frac{2\alpha}{\beta} < 248 \quad 140 < (n+1)^3 < 281$

$n+1 = 6 \quad n = 5$

73. The remainder when 428^{2024} is divided by 21 is_____. **[JEE(Main)April,2024]**

428^{2024} को 21 से विभाजित करने का शेषफल है_____ **[JEE(Main)April,2024]**

Ans. 1

Sol. $21 \times 20 = 420 \Rightarrow (428)^{2024} = (420+8)^{2024}$

$= {}^{2024}C_0\ (420)^{2024} + ...+ {}^{2024}C_{2033}\ (420)^{2023} + 8^{2024}$

$=420$ [some integer] $+ (64)^{1012}$; $63 = 21 \times 3$

$=21 \times 20$ [some integer] $+ (63+1)^{1012}$

$=21 \times 20$ [some integer] $+ 63$[some other integer] $+ 1$

Hence remainder is 1.